

MANIPAL UNIVERSITY JAIPUR
Department of Artificial Intelligence and Machine Learning
Course Hand-out

A. Basic Details:

Programme Name:	B. Tech. CSE (AIML)
Course Name:	Computer Networks
Course Code:	AIM3120
LTPC (<i>Lecture Tutorial Practical Credits</i>):	3 1 0 4
Session:	July 2026– Nov 2026
Class:	V
Course Coordinator:	Dr. Amit Kumar Bairwa
Course Instructor(s):	Dr. Amit Kumar Bairwa, Dr. Hemlata Parmar, Dr. Rahul Sharma, Dr. Rishav Dubey, Dr. Uddalak Chatterjee, Dr. Varun Tiwari, Dr. Vivek Sikarwar, Mrs. Nisha Kundu
Additional Practitioner(s) – if any (<i>Industry Fellow/ Visiting Faculty/ Adjunct Faculty, etc.</i>):	To be identified and appointed later/ or at the beginning of the semester

B. Introduction: The main objective of this course is to familiarize students with computer networks of today which are based on the TCP/IP model and its layered structure.

C. Course Outcomes: At the end of the course, students will be able to

CO Statement	CO	Bloom's Level	Target Attainment %	Target Attainment level
Explain basic concepts, OSI reference model, services and role of each layer of OSI model and TCP/IP, networks devices and transmission media, Analog and digital data transmission.	1	2	100%	3
Develop skills for error detection and correction techniques in the data transmission. Identify Data	2	3	100%	3

framing, flow, and error control protocols at Data Link Layer				
Apply the Internet control protocols – IPv4, IPV6, class full addressing, sub netting and classless addressing.	3	3	100%	3
Analyze the Transport Layer and Its protocols.	4	4	100%	3
Explain the Application Layer, its protocols.	5	2	100%	3

Here, the level indicates the cognitive level of Revised Bloom's Taxonomy.

Information about attainment levels:

Attainment (%)	Level
< 60 %	0
≥ 60% < 70%	1
≥ 70% < 80%	2
≥ 80	3

D. Program Outcomes and Program Specific Outcomes: Mention program outcomes and program specific outcomes of the program.

[PO.1] Engineering Knowledge: Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering to the solution of complex engineering problems.

[PO.2] Problem Analysis: Identify, formulate, research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.

[PO.3] Design/development of solutions: Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.

[PO.4] Conduct investigations of complex problems: Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.

[PO.5] Modern tool usage: Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modelling to complex engineering activities with an understanding of the limitations.

[PO.6] The engineer and society: Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal, and cultural issues and the consequent responsibilities relevant to the professional engineering practice.

[PO.7] Environment and sustainability: Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.

[PO.8] Ethics: Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practices.

[PO.9] Individual and Teamwork: Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.

[PO.10] Communication: Communicate effectively on complex engineering activities with the engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.

[PO.11] Project Management and Finance: Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.

[PO.12] Life-long Learning: Recognize the need for and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.

PROGRAM SPECIFIC OUTCOMES:

[PSO.1] Graduates will be able to examine the applications of Artificial Intelligence and Machine Learning in real-life problems.

[PSO.2] Graduates will be able to design and implement intelligent systems for multidisciplinary problems.

E. Assessment Plan: Clearly write the criteria, its description, and associated marks for assessment of student achievements. In addition, the attendance requirements, assignments need to be mentioned in this section.

Criteria	Description	Maximum Marks
Internal Assessment (Summative)	Sessional Exam I	30
	3 Quiz (10 Marks), Assignment +Viva/Presentation (5+5 Marks), 1 CISCO Certificate (10 Marks) (Accumulated and Averaged) <u>CISCO Certificate</u> Networking Essentials	30
End Term Exam (Summative)	End Term Exam (Close Book)	40
	Total	100

F. Syllabus:

Introduction to Computer Networking Concepts: Data Communications and Networking For Today's Enterprise, Layered Network Protocol Architectures OSI and TCP.

Physical Layer: Data Transmission: Concepts and Terminology, Analog and Digital Data Transmission, Channel capacity.

Transmission Media: Guided and Wireless Transmission, Wireless Propagation, Line-of-Sight Transmission. Signal Encoding Techniques: Analog and Digital Signals.

Digital-To-Digital Conversion: Line Coding Schemes, Block Coding, Scrambling, Analog-To-Digital Conversion: Pulse Code Modulation, Delta Modulation.

Data Link Layer and Logical Link Control (LLC) sub-layer: Asynchronous and Synchronous Transmission, types of Errors. Error Detection Error Correction, Framing for Reliable transmission and Automatic Repeat Request: (ARQ), Stop-and-Wait Protocol, Go-back-N, Selective Repeat. Performance analysis of ARQ protocols. Example protocols such as HDLC and PPP.

Medium Access Control (MAC) sub-layer: Shared media systems: Bus, Star and Ring topologies; TDMA, FDMA, CSMA, CSMA/CD, Ethernet, CSMA/CA protocols; Shared and Switched Ethernet; Related protocols such as ICMP, NAT, ARP and RARP.

Network Layer: IPv4 Addresses and Header, Classfull addressing, Classless addressing, Subnetting, CIDR—Classless Interdomain Routing, VLSM, IPv6 addressing and headers; Routing protocols including distance-vector and link-state approaches.

Transport Layer: Reliable end-to-end transmission protocols; UDP header; Details of TCP header and operation including options headers and congestion control.

Application Layer: Socket Interface and Socket programming; Example protocols such as DNS, SMTP, FTP, and HTTP.

TEXTBOOKS:

1. W. Stallings, Data & Computer Communications (9e), Pearson Education Inc., Noida, 2017.
2. A S Tanenbaum, Computer Networks, 5th Ed., Pearson, 2010.
3. B.A. Forouzan, TCP/IP Protocol Suite, 4th Ed., TMH, 2010.

REFERENCES:

1. J.F. Kurose and K.W. Ross, Computer Networking - A top-down approach, (7e), Pearson, 2017.
2. A. S. Tanenbaum, Computer Networks, (5e), Pearson Education India, 2013.
3. L. L. Peterson and B.S. Davie, Computer Networks- A Systems approach, (5e), Elsevier, 2016
4. B. A. Forouzan and F.Mosharraf, Computer Networks A Top-Down Approach, Mc-Graw Hill, 2017
5. NPTEL Course: NOC: Computer Networks and Internet Protocol, IIT Kharagpur

G. Lecture Plan:

Le ct.	Topics	Session Outcome	Correspo nding CO	Mode of delivery	Mode of Assessing CO
1	Introduction to Course, Course Outcome. Introduction to Computer Networking Concepts: Data Communications and Networking For Today's Enterprise,	To acquaint and clear teacher's expectations and understand students' expectations.	AIM3120.1	Lecture	NA
2	Layered Network Protocol Architectures OSI	Recognize layering architecture and OSI model in Internet	AIM3120.1	Lecture	MTE, Quiz & ETE
3	TCP/IP – Layers and Functioning	Illustrate TCP/IP model	AIM3120.1	Lecture	MTE, Quiz & ETE
4	Physical Layer: Data Transmission: Concepts and Terminology, Composite signal, Bandwidth, Digital Signal: Bit Rate, Bit Length, Digital signal as a composite, Analog Signal, Tutorial	Analyze composite signal	AIM3120.1	Lecture	MTE, Quiz & ETE
5	Transmission Impairment: Attenuation, Distortion, Noise	Identify transmission impairment	AIM3120.1	Lecture	MTE, Quiz & ETE
6	Performance: Bandwidth, Throughput, Latency, Delay, Jitter	Analyze performance in communication	AIM3120.1	Lecture	MTE, Quiz & ETE
7	Transmission Media: Guided and Wireless Transmission, Wireless Propagation, Line-of-Sight Transmission. Introduction, Twisted Pair Cable, Coaxial Cable, Fiber Optic Cable	Identify transmission media	AIM3120.1	Lecture	MTE, Quiz & ETE
8	Digital-to- Analog Conversion: Line Coding Schemes, Block Coding, Tutorial	Illustrate ASK	AIM3120.1	Lecture	MTE, Quiz & ETE
9	Scrambling	HDB3, B8ZS	AIM3120.1	Lecture	MTE, Quiz & ETE
10	Digital-to- Analog Conversion: Digital to Digital	ASK, FSK, PSK, QAM.	AIM3120.1	Lecture	MTE, Quiz & ETE

11	Analog-To- Digital Conversion: Pulse Code Modulation, Delta Modulation.	Framing, Error detection and Correction, Switching	AIM3120.1	Lecture	MTE, Quiz & ETE
12	Error Detection And Correction: Introduction, Types of Errors, Redundancy, Detection versus correction, Coding, Tutorial	Illustrate MAC protocol ALOHA	AIM3120.2	Lecture	MTE, Quiz & ETE
13	Cyclic Redundancy Check, Polynomials	Illustrate MAC protocol – CSMA, CSMA/CD	AIM3120.2	Lecture	MTE, Quiz & ETE
14	Error Correction: Block Code Principles	Illustrate MAC protocol – CSMA/CA	AIM3120.2	Lecture	MTE, Quiz & ETE
15	Flow Control: Stop-and-Wait Protocol, Piggybacking, Sliding Window	Analyze error types	AIM3120.2	Lecture	MTE, Quiz & ETE
16	Error Control: Stop-and-wait ARQ, Go-back- N ARQ, SR ARQ, Tutorial	Apply CRC and error correction mechanism	AIM3120.2	Lecture	MTE, Quiz & ETE
17	HDLC, Configurations and Transfer Modes, Framing	Apply checksum	AIM3120.2	Lecture	MTE, Quiz & ETE
18	Medium Access Control (MAC) sub-layer: Shared media systems: Bus, Star and Ring topologies; ALOHA	Identify flow control in data link layer	AIM3120.2	Lecture	MTE, Quiz & ETE
19	CSMA, CSMA/CD	Identify error control in data link layer	AIM3120.2	Lecture	MTE, Quiz & ETE
20	CSMA/CA, Tutorial	Summarize HDLC and framing	AIM3120.2	Lecture	MTE, Quiz & ETE
21	Network Layer: Store-and-Forward Packet Switching, Services Provided to the Transport Layer	Understanding of packet switching and services provided to transport layer	AIM3120.3	Lecture	MTE, Quiz & ETE
22	Implementation of Connectionless Service, Implementation of Connection-Oriented Service	learn implementation of connectionless and connection-oriented service	AIM3120.3	Lecture	MTE, Quiz & ETE
23	Network Layer: IPv4 Addresses and Header, Classfull addressing,	Knowledge of IP Address, Understanding of need of subnetting	AIM3120.3	Lecture	MTE, Quiz & ETE

	Classless addressing, Subnetting				
24	CIDR—Classless Interdomain Routing, Tutorial	Understanding of CIDR	AIM3120.3	Lecture	MTE, Quiz & ETE
25	VLSM, IPv6 addressing and headers;	Understanding of VLSM	AIM3120.3	Lecture	MTE, Quiz & ETE
26	Routing: Shortest Path Routing, Flooding	Understanding of shortest path routing algorithm	AIM3120.3	Lecture	MTE, Quiz & ETE
27	Distance Vector Routing,	Understanding of distance vector routing algorithm	AIM3120.3	Lecture	MTE, Quiz & ETE
28	Link State Routing, Hierarchical Routing, Tutorial	Understanding of link state routing protocol and Hierarchical routing	AIM3120.3	Lecture	MTE, Quiz & ETE
29	CIDR—Classless Interdomain Routing	Understanding of CIDR	AIM3120.3	Lecture	MTE, Quiz & ETE
30	Related Protocols: NAT, DHCP	Learn Network address translation and network protocols	AIM3120.3	Lecture	MTE, Quiz & ETE
31	ARP, RARP	Understanding of network protocols	AIM3120.3	Lecture	MTE, Quiz & ETE
32	ICMP, IPV4 header format, Tutorial	Understanding of network protocols	AIM3120.3	Lecture	MTE, Quiz & ETE
33	Fragmentation	Learn concept of fragmentation	AIM3120.3	Lecture	MTE, Quiz & ETE
34	RIP, OSPF, BGP	Understanding of dynamic routing protocols	AIM3120.3	Lecture	MTE, Quiz & ETE
Mid Term Examination – I					
35	General Principles of Congestion Control,	Understanding of congestion principles and prevention	AIM3120.4	Lecture	Quiz & ETE
36 *	Techniques for Achieving Good Quality of Service, Tutorial	Understanding of Techniques for achieving good QoS	AIM3120.4	Lecture	Quiz & ETE
37 *	Introduction to Transport Layer, Transport Service Primitives	Understanding of transport layer and service primitives	AIM3120.4	Lecture	Quiz & ETE
38 *	Elements of Transport Protocols, Addressing,	Understanding of elements of transport protocols	AIM3120.4	Lecture	Quiz & ETE

39 *	Connection Establishment, Connection Release	Understanding of connection establishment and release process	AIM3120.4	Lecture	Quiz & ETE
40 *	Flow Control and Buffering, Tutorial	Understanding of flow control and buffering in transport layer	AIM3120.4	Lecture	Quiz & ETE
41 *	UDP, UDP Header	Understanding of UDP	AIM3120.4	Lecture	Quiz & ETE
42 *	TCP Service Model, TCP Protocol	Understanding of TCP	AIM3120.4	Lecture	Quiz & ETE
43 *	TCP Segment Header,	Understanding of TCP segment header	AIM3120.4	Lecture	Quiz & ETE
44 *	Connection Control and Time Control, Tutorial	Understanding of Connection control	AIM3120.4	Lecture	Quiz & ETE
45 *	Introduction to Application Layer	Understanding of application layer	AIM3120.5	Lecture	Quiz & ETE
46 *	DNS—The Domain Name System	Understanding of DNS	AIM3120.5	Lecture	Quiz & ETE
47 *	SMTP, POP	Understanding of email	AIM3120.5	Lecture	Quiz & ETE
48 *	IMAP, MIME, Tutorial	understanding of email	AIM3120.5	Lecture	Quiz & ETE
49 *	HTTP	Understanding of web and protocol	AIM3120.5	Lecture	Quiz & ETE
50 *	SNMP	Understanding of network management protocol	AIM3120.5	Lecture	Quiz & ETE
End Term Examination					

Important Update:

1. As per NAAC Reforms 2024- Binary Accreditation Stakeholder Consultation (University), the mode of delivery should include the followings:

- | | |
|---|--------------------------------------|
| 1. Lecture based teaching-learning | 7. Expeditionary learning |
| 2. Group- teaching and learning | 8. Technology based learning |
| 3. Individual learning/ self-study | 9. Peer teaching |
| 4. Inquiry based learning | 10. Learning through problem-solving |
| 5. Kinaesthetic learning | 11. Flipped Classroom, etc. |
| 6. Game Based learning | |

Course coordinators should include at least 5 pedagogical approaches from the pedagogical approaches as specified above and mention the same in respective lecture types / Mode of delivery.

2. In addition, some lectures should be marked as * that indicates the lectures to be covered by additional practitioners (Industry Fellow/ Visiting Faculty/ Adjunct Faculty, etc.)

H. Course Articulation Matrix:

CO	Statement	Correlation with Program Outcomes														
		PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
1	Explain basic concepts, OSI reference model, services and role of each layer of OSI model and TCP/IP, networks devices and transmission media, Analog and digital data transmission.	3	3	1	3	2				1		3	3	1	2	X
2	Develop skills for error detection and correction techniques in the data transmission. Identify Data framing, flow, and error control protocols at Data Link Layer	3	3	3	3							2	3	1	1	X
3	Apply the Internet control protocols – IPv4, IPV6, class full addressing, sub netting and classless addressing.	3	3	3	3	3				2	2	3	3	1	3	X
4	Analyze the Transport Layer and Its protocols.	3	2	2	3	3				2	2	3	3	1	2	X
5	Explain the Application Layer, its protocols.	3	1	2	1	3			3		2	3	3	1	3	X

1: Low Correlation 2: Moderate Correlation 3: Substantial Correlation